

Short-term effects of fen peatland restoration through the moss layer transfer technique on the soil CO₂ and CH₄ efflux

Cristina Lazcano^{a,b,*}, Cameron Robinson^a, Golnoush Hassanpour^a, Maria Strack^{a,c}

^a Department of Geography, University of Calgary, AB, Canada

^b Department of Natural Resource Management and Environmental Sciences, California Polytechnic State University, San Luis Obispo, CA, USA

^c Department of Geography and Environmental Management, University of Waterloo, ON, Canada

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ABSTRACT

Northern peatlands store approximately one third of the global soil carbon (C) stocks. Peat extraction for horticulture changes C fluxes turning these soils from sinks into large sources of C emitted to the atmosphere as greenhouse gases. Restoring hydrological conditions may not be enough for returning the C sink function of these ecosystems and a recovery of the original Sphagnum-dominated vegetation cover could be necessary. Here we tested the effects of the moss layer-transfer technique (MLTT) on the CO₂ and CH₄ fluxes three years after restoration of a minerotrophic cutover peatland in Quebec, Canada. Gas emissions were compared to nearby undisturbed plots, and plots with restored hydrology but without vegetation transfer. The main drivers for the observed emissions were inferred from the relationships with relevant environmental variables (vegetation cover, water table depth and temperature). Restoration of the extracted minerotrophic fen through the MLTT produced significant changes in the exchange of CO₂, bringing these fluxes closer to undisturbed plots than plots with restored hydrology only, which remained C sources throughout the study. Methane emissions were generally low across all the restored plots and far from natural levels, with only a slight increase at the MLTT restored plots observed during the third year. Even though the MLTT failed to recover the original bryophyte-dominated vegetation, the observed changes in the emissions of CO₂ and CH₄ at the restored plots were strongly associated with changes in vegetation cover (i.e. increase in vascular plants), supporting the use of this restoration technique to recover the C sink function of harvested minerotrophic peatlands.

1. Introduction

Peatland ecosystems constitute one of the largest terrestrial sinks of soil carbon (C) and play a key role in climate regulation with soil carbon stores of approximately 550 Pg of C globally, and over 400 Pg in northern peatlands (Loisel et al., 2014), equivalent to one third of the total terrestrial soil C stocks (Frolking and Roulet, 2007). In these ecosystems storage of C is possible because of the combination of eco-hydrological factors such as the prevalence of anoxic waterlogged conditions, and the presence of peat-accumulating plant species, that slow down decomposition rates. As a result, the dominant process driving C accumulation is the primary production by peat-accumulating vegetation, while C outputs through decomposition remain lower (Vitt, 2006).

Peat extraction for horticulture dramatically changes ecohydrological conditions having a direct impact on C fluxes. In Canada,

most peat is extracted using the vacuum-harvesting technique, where the peatland is first drained through the construction of ditches, and vegetation is removed for the subsequent extraction of the peat using a tractor-pulled vacuum. Vegetation removal virtually eliminates the C inputs, and aeration of the peat profile after drainage increases decomposition rates, which results in the annual release of massive amounts of CO₂ (300–400 g C m⁻²; Waddington and McNeil 2002). Thus, peat extraction transforms peatlands from sinks into net sources of C, contributing to global warming (Strack and Waddington, 2012; IPCC, 2014).

After extraction, spontaneous revegetation of the peatland might occur, particularly in minerotrophic environments, increasing photosynthetic C uptake and CH₄ efflux (Graf and Rochefort, 2009; Mahmood and Strack, 2011) potentially bringing these ecosystems to their original C sequestration function in the long term. However, CO₂ and CH₄ fluxes largely depend on the species of plants spontaneously

* Corresponding author at: Natural Resources Management & Environmental Sciences, California Polytechnic State University, One Grand Ave, 180-514, San Luis Obispo, CA 93407, USA.

E-mail address: lazcano@calpoly.edu (C. Lazcano).

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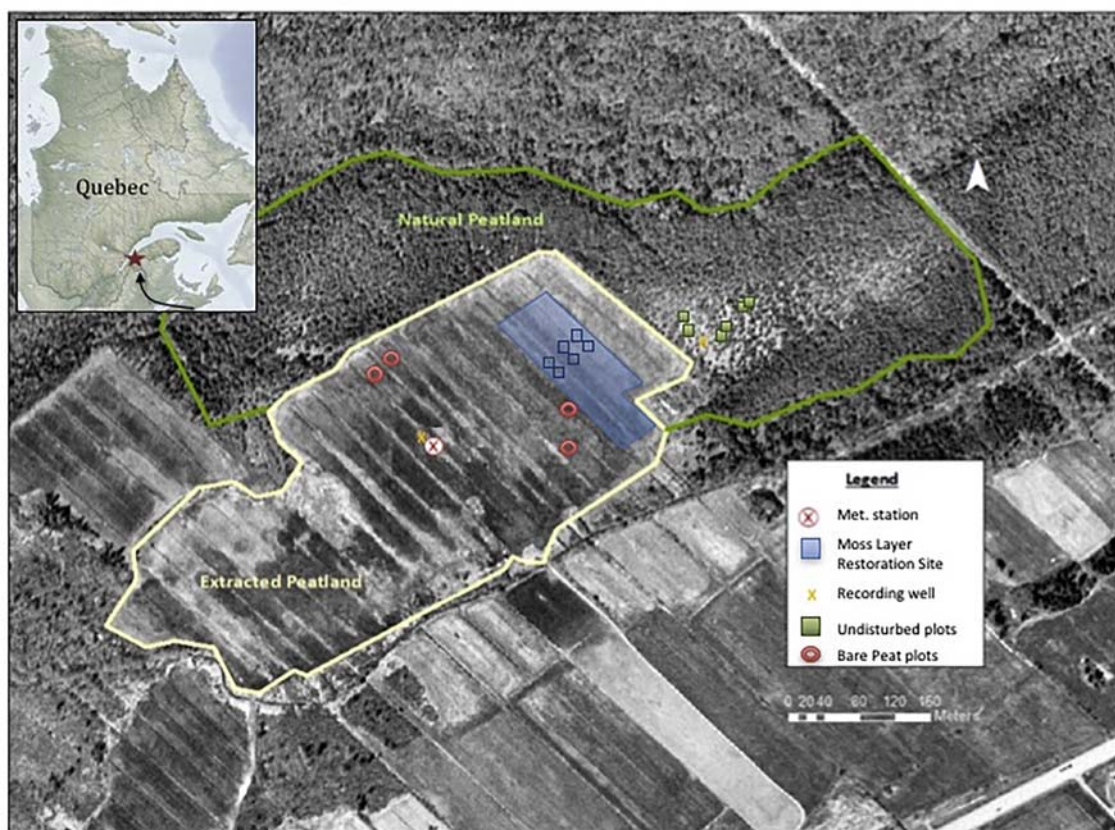


Fig. 1. Overview of the study site and collar location within the extracted and natural areas at the Bic-St. Fabien peatland (Quebec, Canada).

recolonizing the peatland, and certain plants might increase CH_4 fluxes well beyond those observed for natural peatlands (Mahmood and Strack, 2011). Thus, if no restoration is put in place, peatlands will most likely remain net sources of C (Strack and Waddington, 2012). Moreover, spontaneous revegetation alone is unlikely to result in plant communities similar to undisturbed, reference peatlands as bryophyte cover remains lower post-extraction (Graf and Rochefort, 2008a,b).

Fens are minerotrophic peatlands that receive water from precipitation, surface and/or groundwater, as opposed to ombrotrophic peatlands or bogs that receive only water from precipitation. This leads to differences not only in peat chemistry, as some fens can have a higher pH and cation content, but also in plant composition with a higher abundance of vascular plants and brown mosses in fens (Charman, 2002; Vitt, 2006). Bogs may also be transformed into fens after extraction of the peat and exposure of the underlying minerotrophic peat (Graf et al., 2012; Bérubé et al., 2017).

Even though extensive research on fen restoration has been carried out in Europe, this has little applicability in North America because of the different restoration priorities and challenges (Graf and Rochefort, 2008a,b; Lamers et al., 2015). The main challenges for restoration in Europe are the scarcity of pristine fens and eutrophication because of agricultural use, which hampers the collection and establishment of the original plant community. So the goal is to achieve a semi-natural fen state that will allow recovery of biodiversity and preservation of certain endangered plant species (Ehrenfeld, 2000; Lamers et al., 2015). In North America, where one of the major causes of peatland degradation is peat extraction, the main challenges for restoration are the absence of a seed bank and the recovery of hydrological conditions, while the focus is on the recovery of the ecosystem functions such as C sequestration (Rochefort et al., 2003). Here, the existence of larger extent of non-degraded fens, allows for the collection of representative original plant material to be used as diaspores and transferred into degraded areas.

Previous fen restoration experiences in North America have prioritized the vascular plant component because of their larger and faster biomass accumulation potential (Cooper and MacDonald, 2000). However, because of their low decomposition rates and plant structure, mosses also play a key role in peat accumulation, structure and hydrological conditions (Graf and Rochefort, 2009). The moss layer-transfer technique (MLTT) approach to peatland restoration (Quinty and Rochefort, 2003) aims at recovering the peat accumulation and C sink function of bog ecosystems by re-establishing the original Sphagnum-dominated vegetation cover. This involves the introduction of donor vegetation material from a nearby-undisturbed site, protection of the introduced moss propagules with straw mulch and application of a light dose of phosphate fertilizer (Rochefort et al., 2003). Because of the key role of water availability and peat moisture for Sphagnum growth, reclamation plans include necessarily the restoration of hydrological conditions of the peatland by blocking the drainage ditches to raise the water table (Rochefort et al., 2003).

The MLTT has been successfully applied in the restoration of ombrotrophic peatlands, or bogs, where it has shown to have significant effects in C fluxes, efficiently shifting them from growing season C sources into sinks shortly after restoration (Waddington et al., 2010; Strack et al., 2014). However, research on the efficiency of this technique for fen restoration is still scarce. While it has been observed that it can be effective in recovering the peat-forming vegetation layer in extracted fens (Graf and Rochefort, 2008a,b), albeit with low bryophyte cover when applied mechanically (Rochefort et al., 2016), it is not yet clear whether this is accompanied by a recovery of the C sink function and what the impact of this restoration technique is on peat gas exchange to the atmosphere. Therefore, the objective of this study was to assess the effects of MLTT restoration on the fluxes of CO_2 and CH_4 in the short term, three years after restoration of a minerotrophic cutover peatland in Quebec, Canada. Furthermore, we aimed to understand the

main environmental controls on these fluxes. The specific hypotheses to be tested were: (i) that fen restoration using MLTT has a significant short-term impact on the CO₂ flux, reducing emissions and increasing photosynthetic uptake from the atmosphere; (ii) restoration increases CH₄ emissions to levels similar to natural conditions; and (iii) changes in CO₂ and CH₄ fluxes can be explained through changes in water table depth and vegetation abundance.

2. Material and methods

2.1. Study site

This study was conducted at the Bic-Saint-Fabien (BSF) peatland (48°18'N, 68°52'W), 25 km away from Rimouski, Quebec, Canada (Fig. 1). The undisturbed portion of the site is a moderately rich fen, with sparse forest cover and an average peat depth of 4.5 m. The natural vegetation is dominated by Eastern White Cedar (*Thuja occidentalis*) in the forested part, with the moss layer dominated by brown mosses including *Campyllum stellatum*, *Drepanocladus* spp., and *Tomenthypnum nitens*. Peat extraction started at this site in 1946 and continued until 2000, when it was abandoned. To date, a total of 22 ha has been extracted. The area extracted was originally a bog, but extraction of surface layers has exposed peat layers indicative of a poor fen (Bérubé et al., 2017). Restoration activities on this area started in 2008 and were completed in fall 2009. The purpose was to create favorable conditions for the development of a moss layer and the recovery of the C sink function. Briefly, restoration included modification of the extracted peatland hydrology through the blocking of the drainage ditches and construction of dykes to retain snowmelt and raise the water table. In addition, moss diaspores mechanically harvested from a nearby moderate-rich riverine fen were spread in a 1:10 ratio (1 ha of collected material over 10 ha of restored area), and subsequently covered with straw mulch (see Rochefort et al., 2016 for more details).

Six sampling plots were established in the restored site (restored) arranged in three groups of two plots each along three original neighboring peat fields. There is a general slope to the peatland and thus these plots were arranged to capture the gradient in hydrologic conditions along this slope. Another set of six sampling plots was established in the nearby natural undisturbed peatland, three of them on hummocks (undisturbed hummock) and three on hollows (undisturbed hollows). Additionally, in order to quantify the effects of vegetation cover after restoration on CO₂ and CH₄ baseline fluxes, four plots where bare peat was exposed were monitored at the restored area (restored bare). Carbon fluxes in the different plots were monitored from early May until late August in 2010 and 2011 and from early May until early September 2012. Fluxes were assessed biweekly in 2010 and weekly in 2011 and 2012.

2.2. Assessment of CO₂ exchanges

Net ecosystem exchange of CO₂ (NEE) was determined at each plot through the closed chamber method. Chambers consisted of a square collar (60 cm × 60 cm) left in place throughout the study period plus a 60 cm × 60 cm × 30 cm transparent acrylic lid, except at the bare peat site, where chambers consisted of a round (30 cm diameter) collars with a groove for water sealing. Chambers placed on these round collars were 30 cm high. The sleeve of the collars extended 20 cm in soil, which is usually below the rooting zone (Laiho and Finér, 1996). Total chamber height from the soil surface ranged between 48 and 30 cm (depending on the depth of collar insertion) and was used to estimate the exact air volume within each chamber. CO₂ concentration in the chamber headspace was monitored at 15 s intervals for 105 s with a portable infrared gas analyzer (EGM-4, PP Systems, Massachusetts, USA). A battery operated fan mixed the headspace during flux measurement. Flux was determined as the linear change in CO₂ concentration over time correcting for chamber volume and ambient

temperature as recorded with a thermocouple inserted into the chamber. Together with flux measurement, photosynthetically active radiation (PAR) was measured with a quantum sensor connected to the infrared gas analyzer. NEE was measured under a variety of light levels created using shades and ecosystem respiration (ER) determined under an opaque tarp and it is expressed in positive values to indicate CO₂ emissions to the atmosphere. Gross ecosystem photosynthesis (GEP) was calculated as the difference between NEE and ER and is expressed in negative values to indicate CO₂ influx from the atmosphere into the vegetation. At the bare peat plots, only ER was measured, as no vegetation was present.

2.3. Assessment of CH₄ fluxes

Methane fluxes were assessed in the same collars by using an opaque 60 cm × 60 cm × 30 cm chamber equipped with a battery-operated fan to mix the headspace air. Headspace gas samples were collected at regular time intervals (7, 15, 25 and 35 min) after chamber deployment onto the collars through a three-way valve sampling port. Samples (20 mL) were stored in 12 mL pre-evacuated Exetainers (Labco Ltd., UK), and further analyzed for CH₄ concentration by using a gas chromatograph equipped with flame ionization detector (Varian 3800, Varian Canada).

Chamber gas concentrations were converted to mass per volume units assuming ideal gas relations using chamber air temperature values. Fluxes (mg CH₄ m⁻² d⁻¹) were calculated from the linear change in chamber CH₄ concentration over time, taking into account chamber volume and soil surface area. Instantaneous fluxes were calculated on a per minute basis and then extrapolated to daily fluxes, assuming that CH₄ flux at the time of sampling was representative of daily average flux. If the pattern of concentration did not consistently increase or decrease over time or jumped suddenly indicating potential ebullition, the flux value was not used except in cases where the slope was not significantly different from zero indicating a non-detectable, low flux. Other than these low fluxes, values were accepted if the R² values of the linear regression between sampling time and sample concentration were larger than 0.75. This resulted in loss of 13, 18 and 14% of the data in 2010, 2011 and 2012 respectively.

2.4. Ancillary environmental measurements

Vegetation volume in the gas sampling collars in 2011 and 2012 growing seasons was estimated using the 'Fuel Rule', a visual obstruction method developed by Davies et al. (2008). Measurements were not made in 2010 due to low plant cover in the first year post-restoration. A statistically significant relationship was found between vegetation volume determined with this method and destructively harvested biomass for the BSF peatland (Strack and Srivastava, 2010). Vegetation volume was measured at different times during the growing season, and the change in vegetation volume according to the day of the year (DOY) fitted through Gaussian models (Riutta et al. 2007a; Strack et al., 2017). In this way, vegetation volume could be estimated for any given day during the growing season according to:

$$VV = VV_{max} \cdot \exp \left[\left(\frac{JD - JD_{max}}{b} \right)^2 \right] \quad (1)$$

where VV_{max} is the maximum vegetation volume during the season, JD is the Julian day (days of a year numbered from 1 to 365), JD_{max} is the timing of maximum vegetation volume (VV_{max}) and b is the width of the Gaussian curve. In addition, a vegetation survey was carried out in August 2013, to visually estimate percentage cover of different vascular plant species and mosses in the collars at the restored site. At this time, the percentage surface occupied by the hummocks and hollows at the undisturbed site was estimated too.

Soil temperature at 5 and 20 cm depth near the collars was

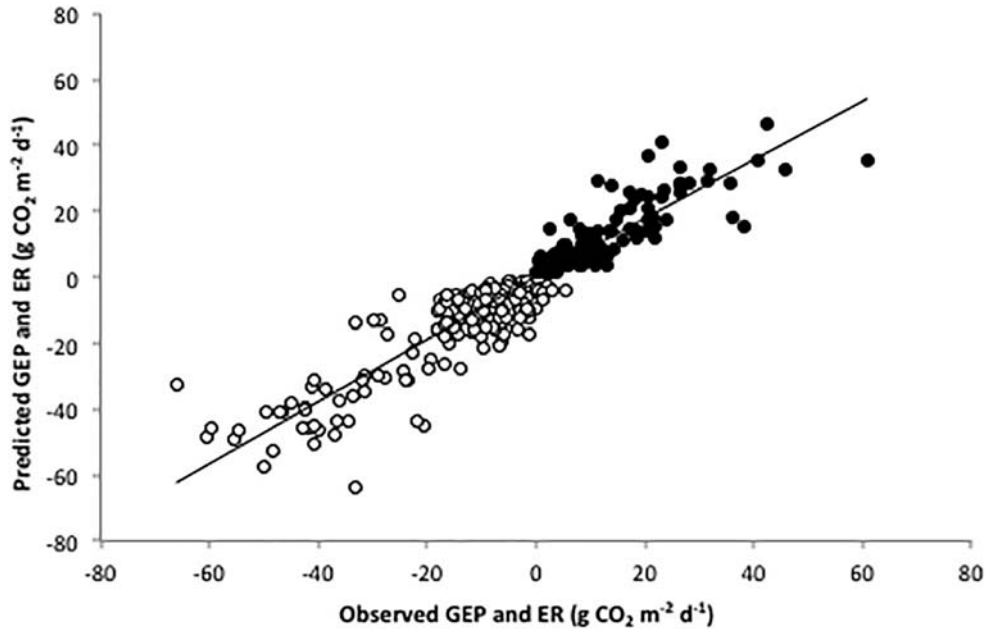


Fig. 2. Linear relationship between observed and predicted GEP and ER values obtained from the seasonal models. Models were able to predict GEP and ER with a goodness of fit (R^2) of 0.71 and 0.70 respectively.

measured during gas flux measurements using a thermocouple. Water table depth was measured in PVC standpipes installed adjacent (within 1 m) to each collar. Continuous data was collected year-round from a meteorological station installed approximately 150 m from the restored plots on the restored site and measured PAR (quantum sensor), precipitation (tipping bucket rain gauge), soil temperature (copper-constantan thermocouples) recorded each minute and averaged every 20–30 min and recorded by a data logger (Campbell Scientific, CR10X). Water level was recorded half-hourly in standpipes located adjacent to the meteorological station (see Fig. 1), in one of the fields near the

construction was based on the existence of significant relationships with GEP. Only variables that increased the explanatory power of the model (R^2) were included. Vegetation volume was included in the models either as a linear (Eq. (2)) or an exponential component (Eq. (3)) whenever a significant linear or non-linear relationship was found, respectively:

$$GEP = \left(\frac{(PAR * P_{MAX})}{(PAR + k)} \right) * e^{\left[-0.5 * \left(\frac{(WL - WL_{OPT})}{WL_{TOT}} \right)^2 \right]} * e^{\left[-0.5 * \left(\frac{(T5 - T_{OPT})}{T_{TOT}} \right)^2 \right]} + (a * VV) \quad (2)$$

$$GEP = \left(\frac{(PAR * P_{MAX})}{(PAR + k)} \right) * e^{\left[-0.5 * \left(\frac{(WL - WL_{OPT})}{WL_{TOT}} \right)^2 \right]} * e^{\left[-0.5 * \left(\frac{(T5 - T_{OPT})}{T_{TOT}} \right)^2 \right]} * e^{\left[-0.5 * \left(\frac{(VV - VV_{OPT})}{VV_{TOT}} \right)^2 \right]} \quad (3)$$

MLTT plots and in the natural fen with pressure transducers (Solinst levellogger), corrected for changes in barometric pressure also logged on site (Solinst barologger).

2.5. Data analyses

All statistical analyses were carried out with SPSS v.23.0 and SAS (SAS Institute, Cary, NC, USA) software programs. Differences in average CH_4 flux, GEP, NEE and ER between treatments (undisturbed hummocks, undisturbed hollows, restored and restored bare) were evaluated using linear mixed models. The interaction between ‘plot’ and ‘DOY’ was entered in the model as random factor whereas ‘treatment’, ‘year’ and the interaction between the two were entered as fixed factors.

Environmental controls (vegetation volume, soil temperature and water table depth) on these variables were also evaluated using Pearson correlations. Linear mixed model and correlation analyses were performed with GEP and NEE flux measurements taken under full sun or no shading conditions (GEP_{max} and NEE_{max}), using a cut off PAR of $1000 \mu mol m^{-2} s^{-1}$ (Bubier et al., 2003).

Cumulative GEP for the study period (May to September) was estimated from measured environmental ancillary parameters by applying modified exponential empirical models (Riutta et al. 2007b, Munir et al. 2014). Selection of ancillary environmental variables for model

where P_{MAX} denotes the potential maximum rate of GEP ($g CO_2 m^{-2} d^{-1}$) not constrained by water level and/or temperature, and k represents the PAR level at which half of the P_{MAX} occurs. The WL represents water level (cm), WL_{OPT} and WL_{TOT} are parameters in a GEP Gaussian response to WL when GEP is at its optimum, and the width of the curve, respectively. The T5 denotes soil temperature at 5 cm depth, T_{OPT} and T_{TOT} are parameters in a GEP Gaussian response to T when GEP is optimum, and the width of the curve, respectively. The VV represents vegetation volume, a is the estimated linear model parameter for the vegetation volume (Eq. (3)); VV_{OPT} and VV_{TOT} are parameters in a GEP Gaussian response to VV when GEP is optimum, and the width of the curve, respectively (Eq. (3)). Model parameters were calculated using non-linear regression with sequential quadratic programming algorithm.

Growing season ER was estimated by using multiple stepwise linear regression model, with soil temperature at 5 cm depth (T5), water table depth (WT) and vegetation volume (VV). Model parameters were fitted by least squares regression.

Due to the differences observed through the mixed model analysis, seasonal GEP and ER models were parameterized separately for each treatment and year. In each case, model parameterization was performed on a randomly selected subset containing 75% of the experimental data whereas the remaining 25% was used for model validation. Model evaluation was based on the existence of: (i) statistically

significant model parameters ($p < 0.05$), (ii) lowest possible standard error of the model parameters and (iii) highest possible adjusted R^2 . The percentage of variability in flux estimates explained by the models in the different treatments and years varied between 22 and 77% (Tables S1 and S2). Yet, overall, models were able to predict GEP and ER with a goodness of fit (R^2) of 0.71 and 0.70 respectively (Fig. 2).

No significant linear relationship between ER, T and WL was found for restored plots in 2010; thus, seasonal total flux was estimated by multiplying the average daily ER values by the number of days of the growing season.

Similarly, for CH_4 flux, no consistent significant linear or non-linear relationship was found with the environmental variables measured at the plots. Therefore, seasonal CH_4 emissions were estimated by multiplying the average daily fluxes by the number of days of the growing season. Seasonal fluxes were also calculated separately for each treatment and year.

3. Results

3.1. Environmental and ecohydrological conditions

Average air temperature during the study period (May to

September) was 15.7 ± 0.4 , 15.3 ± 0.4 and 16.2 ± 0.3 °C, whereas total annual precipitation was 567, 413 and 354 mm in 2010, 2011 and 2012 respectively (Fig. 3, Environment Canada, Rimouski, Quebec).

Soil temperature at 5 cm depth was slightly higher in 2010 and 2012 as compared to 2011 (Year: $F = 21.08$, $p < 0.001$). Significant differences were also found between treatments (Treatment: $F = 3.02$, $p = 0.03$), with restored bare plots having higher soil temperatures than restored plots where the MLTT had been applied (17.5 ± 0.5 °C and 15.6 ± 0.4 °C respectively). Soil temperature at 5 cm depth in hummocks and hollows in the undisturbed peatland was intermediate at 16.8 ± 0.7 °C and 16 ± 0.6 °C, respectively. These differences were consistent in the three years of the study (Treatment*Year: $F = 0.89$, $p = 0.51$, Table 1). A similar trend was observed for soil temperature measured at 20 cm depth in the peat profile, which was significantly higher in 2012 (15.5 ± 0.3 °C) as compared to 2010 and 2011 (12.4 ± 0.5 °C and 11.1 ± 0.3 °C, respectively). Differences between treatments were also found, with bare plots having higher temperatures than restored plots with MLTT (13.9 ± 0.4 °C and 11.9 ± 0.3 °C respectively, Treatment: $F = 4.68$, $p < 0.001$), whereas hummocks and hollows had intermediate values (13.4 ± 0.6 °C and 12.8 ± 0.5 °C, respectively). Soil temperatures at 5 and 20 cm depth were significantly

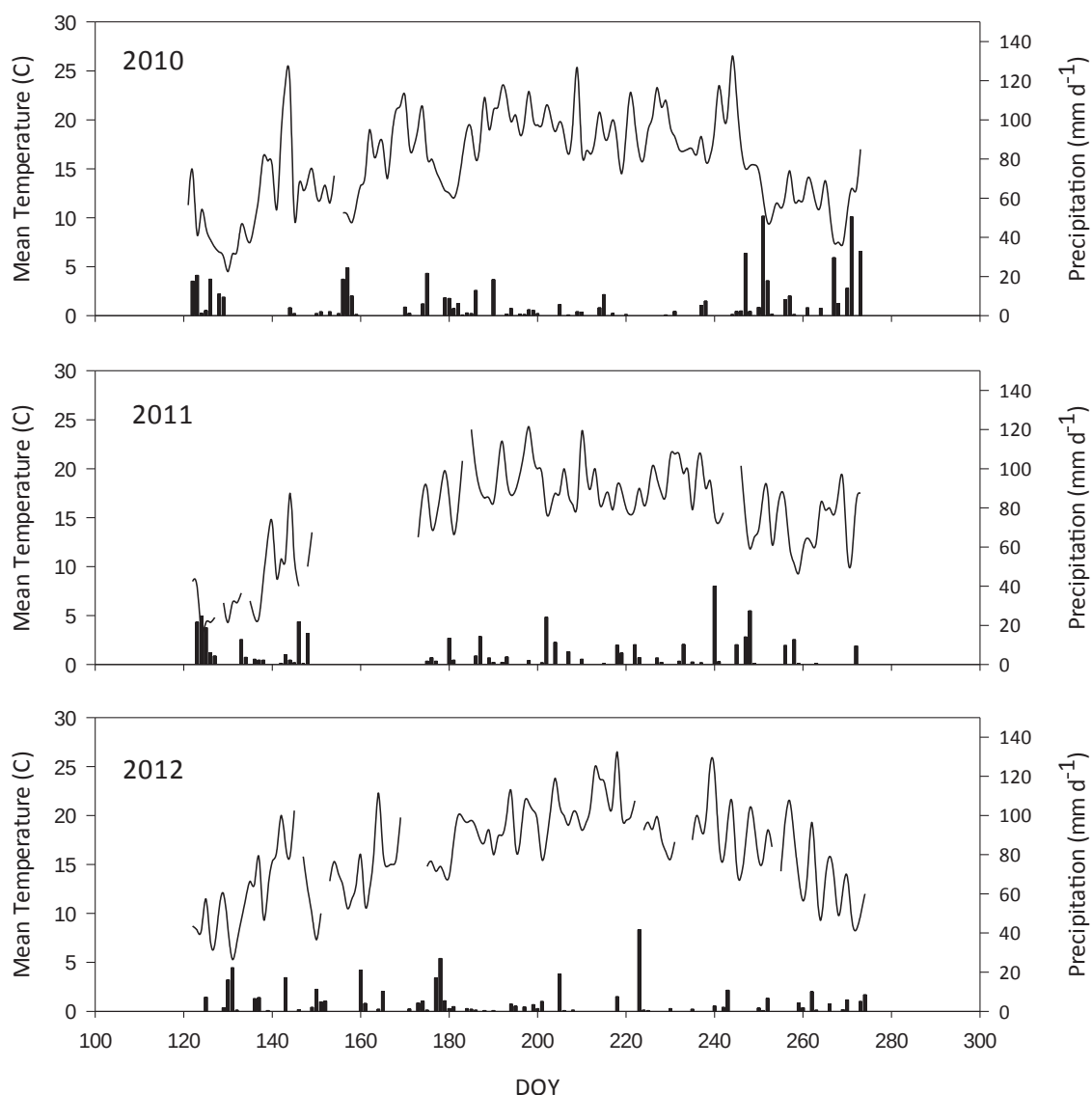


Fig. 3. Changes in mean air temperature and precipitation recorded at the Rimouski meteorological station (Environment Canada) over the course of the study (May to September) in 2010, 2011 and 2012.

Table 1

Average, maximum and minimum water table depth (WT), soil temperature at 5 cm depth (T5), soil temperature at 20 cm depth (T20) and vegetation volume, recorded at the plots over the course of the study. Values are averages $\pm$ s.e.

	WT (cm)	Max WT (cm)	Min WT (cm)	T5 (°C)	T20 (°C)	Vegetation volume
<i>Restored bare</i>						
2010	-34.5 $\pm$ 3.3	-74	-1.3	18.8 $\pm$ 0.9	13.8 $\pm$ 0.6	0
2011	-12.7 $\pm$ 1.1	-30	-1.2	14.9 $\pm$ 0.5	11.7 $\pm$ 0.6	0
2012	-23.4 $\pm$ 1.4	-47.9	-5.5	19.9 $\pm$ 0.5	16.8 $\pm$ 0.4	0
<i>Restored</i>						
2010	-34.2 $\pm$ 2.4	-71.2	-4.1	15.2 $\pm$ 0.5	11.4 $\pm$ 0.3	0
2011	-18.6 $\pm$ 1.4	-48.8	-0.3	13.1 $\pm$ 0.5	9.7 $\pm$ 0.5	3.6 $\pm$ 0.4
2012	-36.8 $\pm$ 2.1	-77.8	-3.8	18.1 $\pm$ 0.6	14.9 $\pm$ 0.5	3.8 $\pm$ 0.5
<i>Undisturbed-hummocks</i>						
2010	-41.2 $\pm$ 2.5	-72.7	-17.5	14.9 $\pm$ 0.7	12.6 $\pm$ 0.4	11.4 $\pm$ 1.7
2011	-30.2 $\pm$ 1.3	-49.7	-13.0	14.8 $\pm$ 1.1	10.0 $\pm$ 0.9	6.41 $\pm$ 0.9
2012	-39.7 $\pm$ 2.1	-69.5	-19.1	17.9 $\pm$ 0.6	16.1 $\pm$ 0.5	6.7 $\pm$ 1.0
<i>Undisturbed-hollows</i>						
2010	-16.8 $\pm$ 2.4	-41.1	-0.6	16.0 $\pm$ 0.7	13.5 $\pm$ 0.4	15.2 $\pm$ 2.7
2011	-6.2 $\pm$ 0.7	-16.7	1.5	13.7 $\pm$ 0.9	10.4 $\pm$ 0.8	11.3 $\pm$ 1.2
2012	-15.0 $\pm$ 1.9	-39	0.4	17.7 $\pm$ 0.9	15.0 $\pm$ 0.6	11.1 $\pm$ 0.9

and positively correlated ($p < 0.001$).

Average water table depth of the plots was deeper in 2010 and 2012 (-30.2 ± 2.3 and -27.7 ± 2.1 cm, respectively) as compared to 2011 (-19.2 ± 2.1 cm) (Year: $F = 100.04$, $p < 0.001$). Significant differences were also observed between the treatments, although these depended on the year considered (Treatment*Year: $F = 10.03$, $p < 0.001$). In 2010, average water table depth was similar in hummocks and both types of restored treatments as compared to hollows where water table position was closer to the soil surface (Table 1). In 2011, water table position at the restored sites was slightly shallower as compared to the previous year, being more similar to the natural hollow sites, whereas in 2012, average water table position of the restored sites decreased again, particularly in the case of restored plots with MLTT (Table 1).

We observed significant changes in the vegetation volume of the plots throughout the study (Treatment*Year: $F = 3.37$, $p < 0.05$). Vegetation volume of the restored plots with MLTT increased significantly with time, reaching similar levels as hummocks by 2012 (Table 1). No vegetation was observed in the bare plots in the three years of the study. Vegetation volume was negatively correlated to water table depth, meaning that water table was lower (more negative values) in plots with higher vegetation volume, and positively

correlated to soil temperature at 5 and 20 cm depth (Table 2).

The vegetation survey carried out in 2013 showed that MLTT restored plots had a percentage cover between 17 and 65% with an average $\pm$ standard deviation of $34 \pm 8\%$. The highest percentage cover was provided by vascular plants with between 12 and 60% cover (average $26 \pm 7\%$), whereas moss cover remained low ($1\text{--}30\%$, average $8 \pm 4\%$). Common vascular plants at these plots were *Sanguisorba canadensis*, *Trichophorum alpinum*, *Carex aquatilis*, *Rhynchospora alba*, *Fragaria virginiana*, *Lycopus uniflorus*, *Eriophorum angustifolium*, *Rubus pubescens* and *Viola* sp. The most abundant moss species was *Campylopus stellatus*, followed by *Sphagnum* spp., *Aulacomnium palustre* and *Tomenthypnum nitens*.

3.2. Carbon dioxide fluxes

Mean growing season GEP_{max} was significantly greater (more negative values) in undisturbed than in restored plots, with more uptake measured in hollows than in hummocks (Treatment: $F = 72.73$, $p < 0.001$; Fig. 4). In addition, undisturbed plots showed larger GEP_{max} values in 2010 as compared to 2011 and 2012, suggesting large interannual variability at these plots. A different trend was observed in restored plots, where GEP_{max} increased significantly in 2012 as compared to 2010 (Treatment*Year: $F = 7.69$, $p < 0.001$; Fig. 4). As no plants were present at the restored bare plots GEP was assumed to be 0.

Ecosystem respiration (ER) was significantly greater on undisturbed hummocks and hollows, followed by restored moss-transfer and bare plots (Treatment: $F = 86.44$, $p < 0.001$; Fig. 4). Respiration rates decreased over the three years of study in undisturbed and restored bare plots, while they increased slightly in the moss-transfer restored plots (Treatment*Year: $F = 13.83$, $p < 0.001$, Fig. 4).

Average NEE_{max} was also significantly different between treatments and years of the study (Treatment*Year: $F = 4.87$, $p < 0.001$; Fig. 4). Negative NEE_{max} values were measured in undisturbed hollows in each of the three years, as well as MLTT restored plots in 2011 and 2012. Positive and similar values were observed in undisturbed hummocks and bare restored plots (Fig. 4).

Throughout the treatments and years, GEP_{max} was strongly correlated with ER (Table 2), meaning that large values of plant photosynthesis and C uptake also corresponded with large respiration rates. Vegetation volume and PAR were strongly correlated to GEP_{max} , ER and NEE_{max} (Table 2). Similarly, a significant positive correlation was found between soil temperature at 5 or 20 cm and ER, and a negative correlation with temperature at both depths and GEP_{max} . Water table depth was negatively correlated with ER and NEE_{max} but positively correlated to GEP_{max} (Table 2).

The largest (more negative) GEP seasonal values were observed in

Table 2

Pearson correlations between average values of the ancillary environmental variables (WT, T5, T20, VV, PAR) and CO_2 exchange (NEE_{max} , GEP_{max} , R_{tot}) recorded at all the plots throughout the three years of the study.

	WT	T5	T20	NEE_{max}	R_{tot}	GEP_{max}	VV	PAR
WT	1	-0.315**	-0.415**	-0.199**	-0.206**	0.022	-0.111*	-0.097*
T5	-0.315**	1	0.859**	0.073	0.254**	-0.132**	0.235**	-0.187**
T20	-0.415**	0.859**	1	0.033	0.267**	-0.166**	0.204**	-0.206**
NEE_{max}	-0.199**	0.073	0.033	1	-0.164**	0.726**	-0.437**	-0.318**
R_{tot}	-0.206**	0.254**	0.267**	-0.164**	1	-0.798**	0.698**	0.373**
GEP_{max}	0.022	-0.132**	-0.166**	0.726**	-0.798**	1	-0.759**	-0.456**
VV	-0.111*	0.235**	0.204**	-0.437**	0.698**	-0.759**	1	0.471**
PAR	-0.097*	-0.187**	-0.206**	-0.318**	0.373**	-0.456**	0.471**	1

**Correlation is significant at the 0.01 level.

*Correlation is significant at the 0.05 level.

WT: water table depth.

T5: soil temperature at 5 cm depth.

T20: soil temperature at 20 cm depth.

VV: vegetation volume.

PAR: photosynthetic active radiation.

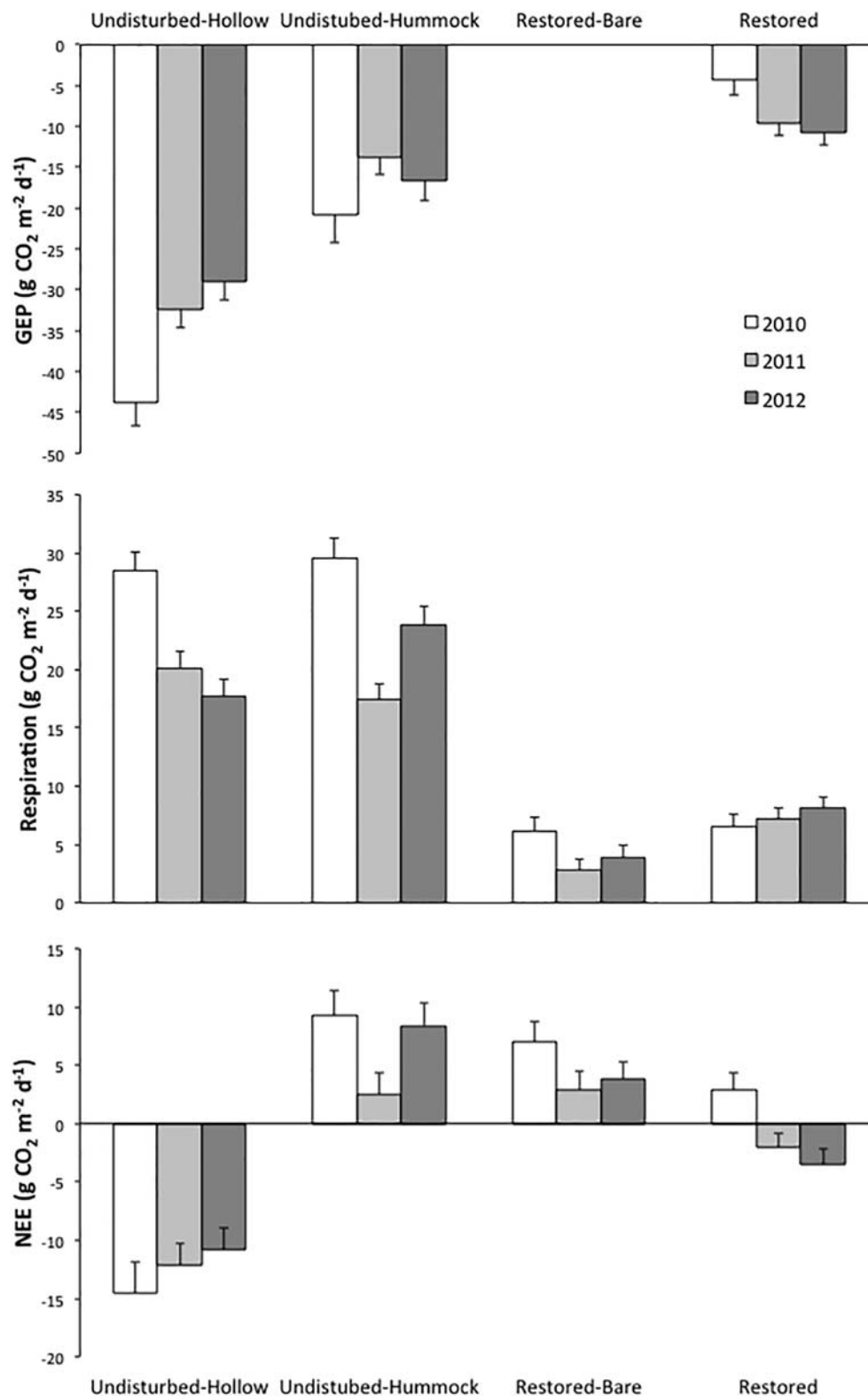


Fig. 4. Average gross ecosystem photosynthesis (GEP), respiration and net ecosystem exchange (NEE) ($\pm$ s.e.) at the undisturbed hummocks and hollows, restored-bare and restored plots where MLTT was applied, in the three years of the study.

undisturbed hollows, followed by undisturbed hummocks and MLTT restored plots (Table 3). Large ER was observed for undisturbed plots (both hummocks and hollows) with values ranging between 396 and 855 $\text{g CO}_2\text{-C m}^{-2}$, whereas lower ER values were found in restored plots (Table 3). Seasonal NEE showed that hollows acted as sources of C in 2010 (due to the high ER rates) but they turned into C sinks in 2011 and

2012 because of the larger uptake through GEP as compared to respiration. On the contrary, in undisturbed hummock's ER exceeded GEP for the three years of the study, and so these microforms acted as net C sources. Taking into account the percent cover of hummocks (24%) and hollows (76%) and the seasonal NEE at each of the microforms, the overall seasonal estimates for the natural site were 373.4, -71.3 and

Table 3

Seasonal (May–September) estimates of the gross ecosystem productivity (GEP), ecosystem respiration (ER), net ecosystem exchange (NEE), CH₄ flux and global carbon balance, for the restored bare, restored MLTT, undisturbed-hummocks and undisturbed hollow plots in 2010, 2011 and 2012.

	GEP (g CO ₂ -C m ⁻²)	se	ER (g CO ₂ -C m ⁻²)	se	NEE (g CO ₂ -C m ⁻²)	se	CH ₄ -C (mg m ⁻²)	se	Global C balance (g m ⁻²)
<i>Restored bare</i>									
2010	0	0	184.6	9.9	184.6	9.9	0.0290	0.02	184.6
2011	0	0	77.9	4.5	77.9	4.5	-0.0001	0.00	77.9
2012	0	0	78.1	8.0	78.1	8.0	-0.0028	0.01	78.1
<i>Restored</i>									
2010	-58.5	11.0	191.4	18.1	132.9	17.8	-0.0941	0.03	132.8
2011	-250.8	21.9	171.5	11.9	-79.3	11.2	-0.0532	0.02	-79.4
2012	-256.1	54.2	191.4	25.5	-64.7	33.2	0.0142	0.05	-64.7
<i>Undisturbed-hummocks</i>									
2010	-279.2	24.0	806.8	72.1	527.5	48.7	-0.0958	0.03	527.4
2011	-242.6	96.6	442.6	8.1	200.1	96.2	-0.0677	0.04	200.0
2012	-224.8	24.9	576.1	67.5	351.3	42.8	0.0292	0.08	351.3
<i>Undisturbed-hollows</i>									
2010	-530.6	45.5	855.1	11.9	324.5	48.5	0.8596	0.46	325.4
2011	-552.5	50.0	396.3	52.3	-156.1	46.2	0.0667	0.03	-156.1
2012	-504.8	42.0	449.6	17.6	-55.2	24.4	0.1661	0.04	-55.1

se: standard error.

41.6 g C m⁻² in 2010, 2011, and 2012 respectively. However, this does not take into account sparse trees on site that would have contributed additional C uptake. Restored bare plots, where GEP was 0 because of the lack of vegetation, remained C sources for the three years, whereas in MLTT plots, NEE was positive in 2010 shifting to negative in 2011 and 2012, resulting in these sites acting as net sinks of C (Table 3).

3.3. Methane flux

We observed large variability in the CH₄ flux, both between treatments and years (Treatment*Year: F = 14.86, p < 0.001). Undisturbed hollows were the largest sources of CH₄ particularly in 2010, although flux decreased significantly in the following years. Methane flux was very low in undisturbed hummocks and restored plots, with negative values in some of the years indicating CH₄ consumption (Fig. 5). Restored MLTT plots increased their CH₄ fluxes slightly over the three years, although no significant differences were detected. Throughout treatments and years CH₄ flux was significantly and positively correlated with vegetation volume (r = 0.22, p < 0.001). Otherwise, no significant correlations were found with water table depth (r = 0.03,

p = 0.45) or soil temperature at 5 or 20 cm depth (r = 0.02, p = 0.62 and r = 0.03, p = 0.41, respectively).

Undisturbed hollows emitted between 66 and 859 mg CH₄-C m⁻² based on seasonal estimates, whereas undisturbed hummocks and restored bare plots had very small fluxes either being a source or a sink of CH₄ depending on the year (Table 3). Restored MLTT plots, consumed relatively large amounts of CH₄ one year after restoration, and became a small source of CH₄ in the years following (Table 3).

4. Discussion

Carbon accumulation in northern peatland ecosystems is mostly driven by low decomposition rates due to the prevalence of waterlogged and anoxic conditions (Limpens et al., 2008). However, results of this study show that restoration of hydrological conditions does not seem to be enough for the recovery of the C sink function in extracted peatlands, highlighting the need for an active restoration of the vegetation cover to increase C sequestration, at least in the short term. Here we observed that, while bare plots were sources of C consistently over the three years of the study, restoration of the extracted minerotrophic



Fig. 5. Average methane flux ($\pm$ s.e.) recorded at the undisturbed hummocks and hollows, restored-bare and restored plots where MLTT was applied, in the three years of the study.

fen through the MLTT produced significant changes in the exchange of CO₂, bringing these fluxes closer to natural conditions than plots with restored peatland hydrology but no revegetation. These changes were observable already three years after the restoration was put in place. At the restored MLTT plots, changes in CO₂ were clearly correlated to the increase in vegetation cover, whereas bare plots remained sources of CO₂ consistently through the three years of the study. Environmental conditions such as water table depth, soil temperature and PAR, had also a strong influence on C fluxes, explaining the variability observed between years, particularly at the undisturbed site. Similar results have been observed across other northern peatlands in Canada where the same restoration technique was applied (Strack et al., 2016).

Methane emissions were generally low across all the restored plots and far from natural levels, with only a slight increase at the MLTT restored plots observed during the third year. In previous studies increases in CH₄ emissions at other restoration sites using the MLTT, have shown to be controlled mainly by WT depth (i.e., hydrological conditions, Strack et al. 2016). Yet, in our study, CH₄ fluxes were more strongly associated to plant cover, with the MLTT restored plots, becoming net sources three years after restoration, when vegetation cover was the highest. Due to the more complex biogeochemical properties of the minerotrophic peat layers as compared to bog peat (Zoltai and Vitt, 1995), usually vascular plants dominate in early stages of secondary succession after restoration, whereas bryophyte establishment is minor (Mahmood and Strack 2011; Rochefort et al., 2016). The presence of vascular plants has shown to increase CH₄ production in anoxic layers of the peat, most likely through the supply of fresh C substrates either through litter deposition or root exudation (Mahmood and Strack, 2011, Bergman et al., 2000, Strack et al., 2017). In addition, aerenchymatous plant species may also increase CH₄ fluxes by venting the CH₄ produced in anoxic peat layers (e.g., Shannon and White, 1994).

It is unclear whether secondary succession will lead the recovery of the original bryophyte-dominated vegetation, even with active restoration (Jones et al., 2018). One of the main issues to assess the success of restoration in the recovery of the original plant community is the scarcity in long term studies on secondary succession of plant communities in degraded peatlands. Alternatively, restoration success can be assessed through the recovery of ecosystem functions such as biogeochemical cycles (Gorham and Rochefort, 2003). Changes in C fluxes observed in this study showed a shift towards natural-undisturbed plots, showing the success of the MLTT technique. As the present study focused on only warm season fluxes, annual C balance of the restored plots was not determined. As non-growing season fluxes are not negligible (Treat et al., 2018), future studies should include wintertime C exchange measurements in restored peatlands to quantify annual fluxes and improve our understanding of C biogeochemistry in cold soils.

The undisturbed sites, including hummocks and hollows, acted as net sources of C in 2010 and 2012, and were weak sinks in 2011, indicating the existence of interannual variability in natural background emission levels from peatlands (e.g., Roulet et al., 2007; Adkinson et al., 2011). Most of the variability in C fluxes was associated to changes in ecosystem respiration and CH₄ efflux, both larger in 2010. In addition, clear differences in ecological functionality and biogeochemical cycling were found between the two microtopographical features studied in the undisturbed site. In the case of hummocks, they were consistently sources of CO₂ resulting in a positive C balance, in spite of their role as a sink of CH₄. Undisturbed hollows were also sources of C in 2010, emitting large amounts of both CO₂ and CH₄. Across microtopographic forms, high respiration rates were responsible for the weak C sequestration capacity of the natural site. These high respiration rates are most likely associated to the presence of trees in the vicinity of the measurement plots (Epron et al., 1999; Hanson et al., 2000). Tree biomass at this site was 8418 g m⁻² and, while our plot-scale fluxes most likely include tree root respiration, they fail to include the GEP of the overstorey vegetation therefore underestimating the C sequestration of the

whole ecosystem (Strack et al., 2016).

5. Conclusion

Restoration of the hydrological conditions did not reduce the C emissions at this minerotrophic peatland at sites that remained unvegetated. The moss layer transfer technique, even though it failed to recover the original bryophyte-dominated vegetation, produced a significant increase in the cover of vascular plants. This increase in vegetation cover caused significant changes in the C fluxes (CO₂ and CH₄) as compared to bare plots, bringing them closer to natural conditions than just restoring the peatland hydrology. More research is needed on the potential of this technique, particularly when donor material is collected and spread mechanically for the recovery of the native bryophyte-dominated vegetation cover found in fens with the associated C cycling, or the effects of alternative, secondary succession pathways on ecosystem-level C fluxes.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ecoleng.2018.10.018>.

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